

BMJ Open Cross-sectional study to investigate the seroprevalence and risk factors of *Toxoplasma gondii* among women attending the antenatal clinic in Namwala, Zambia

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ABSTRACT

Background Toxoplasmosis is a zoonotic parasitic disease caused by *Toxoplasma gondii* (*T. gondii*). It has a wide host range and is capable of vertical transmission in pregnant women, which may lead to undesirable pregnancy outcomes such as congenital malformations, miscarriage, premature birth and stillbirth. This study investigated the seroprevalence of *T. gondii* infection among pregnant women attending the antenatal clinic at Namwala District Hospital in Southern Zambia.

Methods This was a cross-sectional study where blood was collected, and the serum was tested for *Toxoplasma* IgG and IgM. A questionnaire was administered to participants on demographic characteristics and risk factors. Data were entered in Microsoft Excel and exported to STATA version 14 for analysis.

Results A total of 401 women were enrolled in the study from 3 March to 5 August 2021. The seroprevalence of *Toxoplasma* IgG was 4.2% (n=17), while the seroprevalence of *Toxoplasma* IgM was 0.7% (n=3). The median age was 27 (IQR: 24–30) years, and a larger proportion had primary-level education (n=223, 55.6%). The majority (81.6%) of the women were married. None of the risk factors investigated in this study were significant for *T. gondii* infection.

Conclusion There was a low seroprevalence of *T. gondii* infection among pregnant women in the Namwala district of Southern Province, Zambia, and regular screening may not be warranted in this population. Continued research on toxoplasmosis is recommended to understand its epidemiology across Zambia.

INTRODUCTION

Toxoplasmosis is a zoonotic disease caused by the obligate intracellular protozoan parasite *Toxoplasma gondii*.¹ It is estimated to have infected approximately one-third to one-half of the world's population.^{2 3} Humans and other warm-blooded animals are intermediate hosts, while the definitive hosts are felids.⁴ It has a global distribution with prevalence rates ranging from lower than 10% to over 90%.⁵ The global average seroprevalence

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ The study design and population allow the investigation of toxoplasmosis in pregnant women who have the potential of congenitally infecting their unborn children with a possibility of serious clinical implications.
- ⇒ The selection of a rural population in Zambia allows the understanding of potential drivers of *Toxoplasma* exposure in a rural setting not previously investigated.
- ⇒ There is a potential pitfall of this study in extrapolating the findings to other populations as it was done at a single facility, but these findings will be key in designing context-specific interventions for toxoplasmosis.
- ⇒ Risk factor analysis could not be performed due to the relatively low seroprevalence of *Toxoplasma* antibodies in this population.

of latent toxoplasmosis, a condition where a previously infected person is asymptomatic, was estimated to be 33.8%, with the highest seroprevalence in South America at 56.2%.⁶

Humans and animals can become infected by ingesting undercooked meat contaminated with *T. gondii* tissue cysts, direct infection through drinking water and ingesting food contaminated with *T. gondii* oocysts and congenital transmission from an infected mother to the fetus during pregnancy.^{7 8} Other transmission routes have been suggested, including drinking contaminated milk and blood transmission.⁹

Toxoplasmosis is of clinical significance in immunocompromised individuals and pregnant women. Although the infection is generally asymptomatic in individuals with a competent immune system, it can cause undesirable disease sequelae in immunocompromised individuals, such as meningoencephalitis with eye and central nervous

system involvement.¹⁰ Vertical transmission of *T. gondii* to the unborn baby in pregnant women, a condition referred to as congenital toxoplasmosis (CT), can result in miscarriages, premature birth, congenital malformation and stillbirths, with the severity of the undesirable birth outcomes decreasing with increased gestational age.¹¹ The disease may present with clinical complications at birth or later in life, which may include miscarriages, stillbirths and congenital diseases such as neurocognitive deficits and chorioretinitis.¹² The incidence of CT has been reported in France (2.9/10 000 live births),¹³ Netherlands (8.1/10 000 live births) and the USA (0.2/10 000 live births).¹⁴

Treatment with sulphadoxine pyrimethamine against malaria in pregnant women has also been found to have protective effects against the undesirable consequences of *T. gondii* infection.¹⁵ The current management package for pregnant women in Zambia does not include routine screening for toxoplasmosis, as laboratory testing is restricted to HIV, syphilis, haemoglobin and urinalysis.¹⁶ This could be because there is a lack of compelling evidence to inform policy on the need to screen for toxoplasmosis, as there is a paucity of studies in Zambia on the seroprevalence and impact of toxoplasmosis in pregnant women. To our knowledge, only one previous study investigated toxoplasmosis in pregnant women at a large referral hospital in Lusaka, Zambia, and reported a seroprevalence of *Toxoplasma* IgG of 5.9% and an absence of *Toxoplasma* IgM antibodies.¹⁷ This study was carried out in an urban community with a different social ecological environment and possible transmission routes. The lack of extensive studies illustrates the need for more research into the disease to guide policy. There has been an increased interest in toxoplasmosis in rural areas where there is extensive livestock breeding. This provides a risk of *Toxoplasma* oocyst contamination to the animals and further to humans.^{17 18} Here, we carried out a

study in the rural district of Namwala in Southern Zambia among pregnant women attending the antenatal clinic at Namwala District Hospital.

MATERIALS AND METHODS

Study design, area and period

This was a cross-sectional study conducted at Namwala District Hospital in Southern Province, Zambia (figure 1). Namwala covers an expanse of land approximately 10 000 square kilometres and lies between latitudes 15 and 17°S and longitudes 25 and 27°E.¹⁹ It is a rural area whose population is largely composed of pastoral farmers who share common ecological environments with animals, as livestock rearing is the main economic activity in the district.²⁰ Namwala district hospital is a level 1 hospital, which is the largest referral health facility in the district supporting a catchment area that includes rural health centres including Belina, Kalundu, Chikwata, Location and Maunga. The target population was pregnant women attending the antenatal clinic at Namwala District Hospital from 3 March to 5 August 2021

Study aims and population

The aim of this study was to determine the seroprevalence of toxoplasmosis and its risk factors in pregnant women attending Namwala District Hospital. A total of 462 pregnant women were approached to participate in the study, with 401 who finally enrolled, giving a response rate of 86.8%. Pregnant women who presented with clinical conditions requiring urgent attention and those who refused to give consent were excluded from the study.

Sample size and sampling technique

To estimate the required sample size for the seroprevalence study and risk factors, we used the sample size calculation for cross-sectional studies.²¹ We employed a

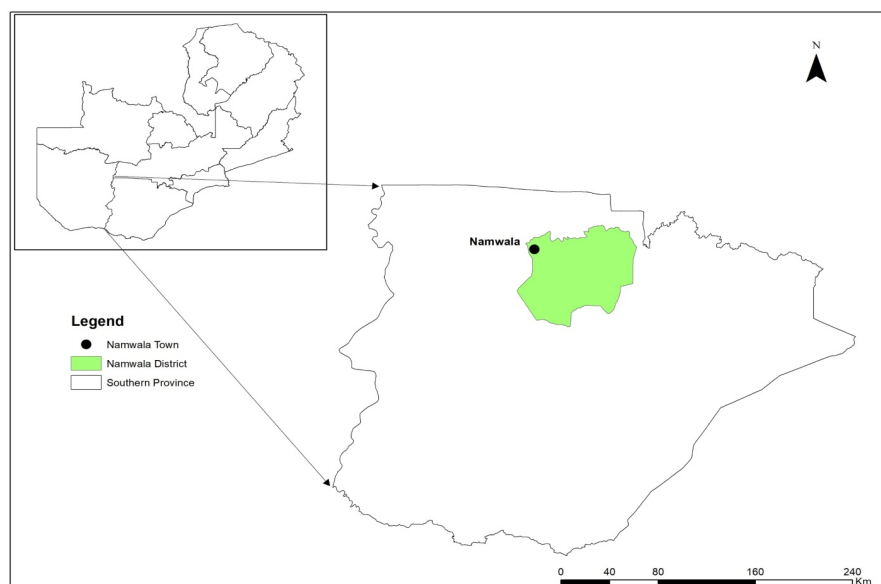


Figure 1 Map of Namwala District in Southern Province, Zambia.

Table 1 *Toxoplasma* seropositivity to IgG and/or IgM and sociodemographic characteristics n=401

Variable	Total population N(%)	Toxoplasmosis status IgG and/or IgM		P value
		Negative n, (%)	Positive n (%)	
Demographic characteristics				
Age median (IQR)	27(24-30)	27(24-30)	26(21-31)	0.079
Gestational age median (IQR)	26(22-28)	26(22-28)	26.5 (24–28)	
Education				
Primary	223 (55.6)	211 (55.3)	12 (63.2)	0.433
Secondary	12 (3.0)	12 (3.1)	0 (0.0)	
Tertiary	10 (2.5)	8 (2.1)	2 (10.5)	
None	156 (38.9)	151 (39.5)	5 (26.3)	0.623
Occupation				
Other*	93 (23.2)	90 (23.6)	3 (15.8)	
Farmer	308 (76.8)	292 (76.4)	16 (84.2)	0.759
Residence				
Belina	87 (21.7)	83 (21.7)	4 (21.1)	
Chikwata	23 (5.7)	23 (6.0)	–	0.045
Kalundu	30 (7.5)	33 (8.6)	2 (10.5)	
Location area	28 (7.0)	25 (6.5)	3 (15.8)	
Maunga	66 (16.5)	63 (16.6)	3 (15.8)	0.005
Other†	162 (40.4)	155 (40.6)	7 (36.8)	
Marital status				
Married	327 (81.6)	311 (81.4)	16 (84.2)	0.757
Unmarried	74 (18.5)	71 (18.6)	3 (15.8)	
Parity				
No child	52 (13.0)	46 (12.0)	6 (31.5)	0.005
One child	256 (63.8)	247 (64.7)	9 (47.4)	
Two and above	93 (23.2)	89 (23.3)	4 (21.1)	
Have you taken IPTp—sulphadoxine pyrimethamine?				0.757
No	46 (11.5)	40 (10.5)	6 (31.6)	
Yes	355 (88.5)	342 (89.5)	13 (68.4)	
Trimester				0.757
1	6 (1.5)	6 (1.6)	0 (0)	
2	250 (62.3)	239 (62.6)	11 (57.9)	
3	145 (36.2)	137 (35.8)	8 (42.1)	
*(Unemployed, Trader, Gardner, Housewife, Maid, Nurse, Policewoman, Teacher).				
†(Banamwanze, Banga, Chaako, Chibunze, Chinyemu, Cooperative, Council compound, Kaambwe, Kaliweza).				
IPTp, Intermittent Preventive Treatment of Malaria for Pregnant Women; IQR, interquartile range.				

* (Unemployed, Trader, Gardner, Housewife, Maid, Nurse, Policewoman, Teacher).

† (Banamwanze, Banga, Chaako, Chibunze, Chinyemu, Cooperative, Council compound, Kaambwe, Kaliweza).

IPTp, Intermittent Preventive Treatment of Malaria for Pregnant Women; IQR, interquartile range.

conservative prevalence of 50% due to the absence of *Toxoplasma* seroprevalence data with the only previous study in Zambia conducted in a geographically dissimilar area.¹⁶ Considering a finite population of 2400 women attending the antenatal clinic at Namwala District Hospital over the study period, a 20% nonresponse and a precision of 5%, we estimated a minimum sample size of 398 participants.

Pregnant women who met the inclusion criteria were approached, and informed consent was obtained.

Approximately 100 women attended the Namwala antenatal clinic weekly. Those who consented were then enrolled using a systematic random sampling approach selecting every fourth participant.

Questionnaire and study measures

Questionnaire administration was conducted by a trained interviewer. The questionnaire had two parts with questions adapted from previous studies.^{6 16 22–24} The first part of the questionnaire (Section A) had questions

Table 2 *Toxoplasma* seropositivity by risk factors n=401

Variable	Total N(%)	Toxoplasmosis status	IgG and/or IgM	P value
		Negative N _i (%)	Positive N(%)	
Risk Factors to toxoplasmosis				
Do you keep a cat?				0.219
No	176 (43.9)	170 (44.5)	6 (31.6)	
Yes	225 (56.1)	212 (55.5)	13 (68.4)	
Do you clean cat litter?				0.204
No	178 (44.4)	172 (45.0)	6 (31.6)	
Yes	223 (55.6)	210 (55.0)	13 (68.24)	
Do you drink unpasteurised milk?				0.299
No	378 (94.3)	359 (94.0)	19(100)	
Yes	23 (5.7)	23 (6.0)	–	
Are you in regular contact with soil such as farming?				0.241
No	8 (2.0)	7 (1.8)	1 (5.3)	
Yes	393 (98.0)	375 (98.2)	18 (94.7)	
Do you handle manure?				0.731
No	17 (4.2)	16 (4.2)	1 (5.3)	
Yes	384 (95.8)	366 (95.8)	18 (94.7)	
Do you boil water before drinking?				0.166
No	367 (91.5)	351 (91.9)	16 (84.2)	
Yes	34 (8.5)	31 (8.1)	3 (15.8)	

on demographics (10 questions), and the second part (Section B) consisted of questions on risk factors for toxoplasmosis, which were adapted from studies conducted in similar settings (eight questions). We performed face validation by distributing the questionnaire to experts in the Public Health Department at the Copperbelt University, School of Medicine. The questionnaire was pretested on 30 women at the Chipulukusu clinic in Ndola, obtaining a Cronbach's alpha of 0.76, which indicated good reliability according to methods described previously.²⁵ The results from the pre-test were used to optimise the questionnaire but were removed from the final analysis.

Blood collection, transportation and analysis

Approximately 4 mL of venous blood was collected in plain containers by a laboratory technologist using aseptic methods. The sample collection was conducted from the Mother and Child Health section of the Hospital and immediately transported to the Namwala District Hospital Laboratory for processing. After clotting, the blood was centrifuged at 3000 revolutions per minute for 3 min, and the serum was aliquoted and stored in cryovials at –20°C. All collected samples were transported monthly under cold storage using SnoMaster mobile cooler boxes (SnoMaster, Johannesburg, South Africa) to the Tropical Diseases Research Centre (TDRC) laboratory in Ndola, Zambia, for analysis. The samples were analysed for *Toxoplasma* IgM and IgG using the immunochromatographic lateral flow test OnSite Toxo IgM/IgG Combo Rapid Test (CTK Biotech Inc, CA, USA) according to the

manufacturer's instructions. This assay has been evaluated against traditional enzyme-linked immunosorbent assay (ELISA) methods and found to give optimal results in the diagnosis of *T. gondii* antibodies.^{26 27} Briefly, approximately 10 µL of the serum was added to the sample well of the immunochromatographic test cassette. Two drops of sample diluent were then added to the diluent well and incubated on a horizontal surface for 12 min. All results were read and interpreted after the incubation period. A valid positive result was interpreted as the presence of a red line on the IgG or IgM in the presence of a C (control) line. The absence of either the IgG or IgM line in the presence of a C line represented a negative result. All *Toxoplasma*-seropositive mothers were referred to the attending physician in the antenatal clinic for further management.

Statistical analysis

Data from questionnaires were entered in Microsoft Excel (Microsoft, Redmond, WA, USA) for cleaning (online supplemental file 1) and then transferred to STATA version 14 (Stata Corp, College Station, Texas, USA) for statistical analysis. We reported the frequency (%) for categorical variables and the median (IQR) for continuous variables. Univariate analysis was done to select covariates to include in the multivariate level at $p < 0.25$. All selected covariates were then included in the multivariate logistic regression model using the enter method. The regression model was not statistically significant ($\chi^2 = 14.033$, $p = 0.121$). The regression model

Table 3 Risk factors for *Toxoplasma* IgG/IgM seropositivity

Covariate	N	aOR (95% CI)	P value
Contact with soil			
No	8	Ref	Ref
Yes	393	4.323 (0.458 to 40.794)	0.201
Boiling water			
No	367	Ref	Ref
Yes	34	0.987 (0.171 to 5.687)	0.989
Education			
Education—uneducated	156	Ref	Ref
Education—primary	223	0.422 (0.137 to 1.303)	0.134
Education—secondary	12	29249032.06 (0.000 -)	0.999
Education—tertiary	10	0.119 (0.013 to 1.128)	0.064
Parity			
Parity no child	52	Ref	Ref
Parity one child	256	4.468 (1.332 to 14.992)	0.015
Parity two or more children	93	3.714 (0.887 to 15.559)	0.073

explained 10.8% (Nagelkerke R^2) of the variance in toxoplasmosis among pregnant women and correctly classified 95.3% of the cases. This implies that the combined effect of environmental factors (keeping cats, cleaning cat litter, contact with soil, boiling water) and human host factors (education, parity) were insufficient for susceptible pregnant women to develop toxoplasmosis in this study.

Ethical approval and consent to participate

Ethical approval for the study protocol (online supplemental file 2) was obtained from the TDRC Research Ethics committee (Institutional Review Board Number 00002911) (online supplemental file 3). Authority to conduct the study was granted by the National Health Research Authority. Permission to conduct the study at Namwala District Hospital was obtained from the office of the Provincial Health Director for Southern Province and District Health Director for Namwala district. Written informed consent was obtained from all participants who were enrolled in the study after informing them about the study, the risks being minimal pain and discomfort at the site of needle puncture during blood collection and their right to withdraw at any given time during the study. All data for the study were restricted to the investigators, and no participant identifiers were used, as all women were assigned a unique study number. The findings of the study were shared with hospital management in a dissemination meeting held after the study. Study identification numbers of participants that were seropositive to *T. gondii* antibodies were referred to routine antenatal care management through the attending clinician who had a register with patient identifiers that were blinded to the investigators.

Patient and public involvement

There was no direct patient and public involvement. The findings from this study were shared with Namwala Hospital Management.

RESULTS

The study enrolled 401 pregnant women with a median age of 27 (IQR: 24–30) years. Table 1 shows details of the cross-tabulation of sociodemographic characteristics and toxoplasmosis detection rate. The largest proportion, 223 (55.6%), attained a primary level of education, 308 (76.8%) were farmers, and 160 (40.4%) resided in other areas (Banamwanze, Banga, Chaako, Chibunze, Chinyemu, Cooperative, Council compound, Kaambwe, Kaliweza). In addition, most women (327, 81.6%) were married, 256 (63.8%) had one child and 355 (88.5%) received Intermittent Preventive Treatment of Malaria for Pregnant Women. Close to two-thirds (250, 62.3%) of the participants were in their second trimester (table 1).

Of the 401 pregnant women, 19 (4.7%, 95% CI: 2.9 to 7.3) were positive for *Toxoplasma* antibodies, 0.7% (n=3) were seropositive for IgM and 4.2% (n=17) were seropositive for IgG, while one of the participants was seropositive for both IgG and IgM. Among those that were seropositive to *Toxoplasma* IgG antibodies, only one had a gestational age of less than 20 weeks.

Slightly above half, 225 (56.1%) owned a cat, and 223 (55.6%) regularly cleaned cat litter. Nearly everyone, 378 (94.3%), drank unpasteurised milk, 392 (97.8%) were in regular contact with soil, 384 (95.8%) handled manure and 367 (91.5%) did not boil drinking water (table 2). All respondents declined eating undercooked or uncooked meat and eating uncooked or undercooked vegetables.

None of the factors that were included in the multivariate were significant as the model was not significant. Table 3 shows the risk factors for *Toxoplasma* IgG/IgM seropositivity.

DISCUSSION

This study is the second study to investigate the seroprevalence and associated risk factors in pregnant women in Zambia and the first in a rural setting such as Namwala. The only previous study was conducted at the University Teaching Hospital, the capital city of Zambia.¹⁶ *T. gondii* remains a disease with wide distribution globally with a heterogeneous reach across and within countries.²⁸

This study found a seroprevalence of 0.7% for *Toxoplasma* IgM and 4% for *Toxoplasma* IgG, and an overall *Toxoplasma* seroprevalence of 4.7% (95% CI: 2.9 to 7.3) in pregnant women in the Namwala district of Zambia, presenting a low endemic picture. The overall seroprevalence is consistent with that reported in a previous study in Zambia with an overall seroprevalence of 5.87% (95% CI: 3.8 to 8.6).^{16 29} Studies in Portugal and Ethiopia have reported higher levels of overall *Toxoplasma* seroprevalence of 24.4% and 81.8, respectively.^{30 31} The presence of IgM antibodies has been associated with poor birth outcomes with IgM appearing first and then receding with IgG becoming more predominant.^{32 33} The low prevalence of *Toxoplasma* IgM is in agreement with those previously reported from Iran and Egypt, which found an IgM seroprevalence of 1.3% and 0.5, respectively.^{34 35} This could suggest a lower risk of unwanted birth outcomes to *Toxoplasma* infection in this population. In contrast to our study, several studies have reported an absence of *Toxoplasma* IgM in pregnant women in Mexico, Tanzania, Ghana and Zambia.^{12 16 36–38}

Toxoplasma IgM seroprevalences higher than those in our study have been reported in several studies, ranging from 2.8% in Columbia to 6.3% in India.^{2 39–43} These differences could be driven by inherent differences in the drivers of infection, such as high consumption of uncooked meat, cat ownership and varying ecosystems and cultures in different countries and areas.⁴⁴ Additionally, the multiplicity of laboratory methods employed by different studies could also account for the differences, as some studies used ELISA methods, while others used immunochromatographic tests and microparticle immunoassays. We reported a seroprevalence of 4% for *Toxoplasma* IgG in the current study. This is comparable to the *Toxoplasma* IgG prevalence reported in South Korea of 4%⁴⁵ and Mexico of 4.2%.³⁸ Slightly higher *Toxoplasma* IgG estimates were seen in Zambia (5.9%) and Mexico (6.1%).^{12 16} In contrast, much higher seroprevalences ranging from 25% to 74.9% have been reported among pregnant women worldwide.^{31 34 36 37 41 43 46–48} These differences could be due to variations in ecologies that drive infection and lower infection rates, sparse population geographies that offer little opportunity for an infection

encounter and possible clustering of cases in areas not yet investigated.

None of the risk factors included in our study significantly explained the seroprevalence of toxoplasmosis in our study.

Our findings cannot be deemed representative of other populations in Zambia and must not be generalised as such. The study initially proposed to investigate the seroprevalence of toxoplasmosis in Namwala, Nyimba and Ndola districts but was limited due to resources and travel restrictions as data collection was conducted at the height of the COVID-19 pandemic. Despite this, our study presents important findings in a rural *Toxoplasma*-susceptible population in Zambia with implications for mitigation using the one-health approach. Our findings must be interpreted with caution as we could not conclusively confirm active infections to augment the findings from the IgG and IgM assays through the use the Avidity assay. Additionally, the OnSite Rapid Diagnostic test has been reported to have low specificity in some populations, which could have a bearing on our findings.

CONCLUSIONS

We found a low seroprevalence of *T. gondii* infection in pregnant women in the Namwala District of Southern Zambia at 4.7%. We recommend deliberate health education in *Toxoplasma* preventive practices to pregnant women during their first antenatal visit to avoid possible primary infection during pregnancy. Screening for *T. gondii* infection may not be warranted due to the low prevalence observed in this population. We recommend more research to understand the epidemiology of *T. gondii* infection across Zambia.

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Contributors VD and AMP conceptualised the study, VD and SDZ collected the data, VD and MM conducted the formal analysis, VD, MM and SDZ did the data curation, VD wrote the initial manuscript and AMP supervised the study. All authors read, reviewed and approved the final version of the manuscript following the STROBE checklist (online supplementary file 4).

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Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, conduct, reporting or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Consent obtained directly from patient(s).

Ethics approval This study involved human participants and was approved by Tropical Diseases Research Centre Institutional Review Board (Institutional Review Board Number 00002911). Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement All data relevant to the study are included in the article or uploaded as supplementary information.

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